

DIAL — Supplementary Material

a direction-invariant audit metric for batch-corrected cancer subtype classifiers

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Overview. This supplementary document collects seven reviewer-facing robustness analyses commissioned for the v5.1 DIAL paper, a complete proof of the label-flip lemma, and a code/data-availability statement. Each section asks: if we change the batch-correction estimator (S2), the DE test (S5), the classifier paradigm (S6), or the correction family (S7), does the thyroid-specific DIAL flip survive? Bottom line: the THCA flip is robust under the count-native ComBat-seq estimator (S2), is the only cohort flagged by random-effects meta-analysis (S3), is a gene-level phenomenon that pathway aggregation eliminates (S4), is not an artefact of the Welch t -test used upstream (S5), persists under a variational quantum classifier (S6), and reshuffles sensibly under FastRNA, BUHMBOX, and LMM Han-lab methods (S7).

S1 Mutation label schemes and exclusions

The cohort availability matrix is given in Table 1 of the main text; here we document only the mutation-label conventions and the exclusion reasons for cohorts not carried through to the DIAL analysis.

Subtype labels by cancer.

- **THCA** — BRAF V600E vs RAS (NRAS/HRAS/KRAS) hotspot mutations. Wild-type and RET/PTC fusion samples are excluded.
- **SKCM** — BRAF vs NRAS hotspot mutations (mutually exclusive in TCGA-SKCM).
- **LGG** — IDH1/IDH2 mutation status (mutated vs wild-type).
- **LUAD** — KRAS vs EGFR mutation status (mutually exclusive when both are called in the same sample, resolved to the higher-VAF call).
- **COAD** — MSI-H vs MSS (mismatch-repair status from TCGA MC3 or GEO companion annotation).

Exclusion reasons. Seven cohorts considered during Phase 1 were excluded: GSE33630, GSE29265 (THCA, no mutation labels); GSE65904 (SKCM, no labels); GSE16011 (LGG, no labels); GSE4271 (LGG, HTTP 404 on GEO FTP); GSE72094 (LUAD, GPL15048 probe-gene mapping failed); GSE17536 (COAD, no MSI/MSS labels).

S2 ComBat-seq robustness

We re-ran the v5.1 LODO DIAL pipeline on THCA with two batch-correction variants: (a) the v5.1 baseline, ComBat (`pycombat_norm`) on log2-TPM; and (b) ComBat-seq [Zhang et al., 2020] via `inmoose.pycombat.pycombat_seq` on raw STAR unstranded counts for the RNA-seq cohort, followed by a cross-platform ComBat step to bridge to the Affymetrix microarray values. ComBat-seq is mathematically defined only on RNA-seq counts; we therefore cannot run it jointly across a count cohort and a microarray cohort, and the hybrid above is the strongest defensible construction given that THCA has exactly one cohort of each type.

Four of five classifiers show $|\Delta\text{DIAL}| < 0.1$: LogReg- ℓ_2 $0.494 \rightarrow 0.495$, LogReg-elasticnet $0.492 \rightarrow 0.494$, GradientBoosting $0.334 \rightarrow 0.273$, XGBoost $0.013 \rightarrow 0.000$. Only Random-Forest shrinks appreciably ($0.324 \rightarrow 0.129$; post-correction AUC $0.177 \rightarrow 0.372$), and its AUC remains well below 0.5. No classifier recovers BRAF/RAS signal above chance under either correction. The flip is therefore not an artefact of log2-TPM handling; switching to the count-native estimator leaves four of five classifiers in the same DIAL regime. (Artefact: `results/v8_statgen/v8_combatseq-vs-combat.tsv`)

S3 Random-effects meta-analysis

Background. The primary analysis measured DIAL for five classifiers on five TCGA-derived cohorts. To formalise that the thyroid signal is specific rather than a heavy-tailed pan-cancer phenomenon, we applied random-effects meta-analysis in the style of METASOFT [Han and Eskin, 2011, 2012, Lee et al., 2017].

Methods. Per-classifier, per-cancer DIAL is the effect size θ_i ($i = 1..5$). $\text{SE}(\text{DIAL})$ is approximated by Hanley–McNeil $\text{SE}(\text{AUC}_{\text{post}})$ using class counts from the harmonisation table; DIAL is a shifted/flipped function of AUC_{post} so this is valid to first order. SE is floored at 10^{-4} to avoid divide-by-zero under perfect separation. Between-study τ^2 uses DerSimonian–Laird; Cochran’s Q is tested against χ^2 with $k - 1 = 4$ d.f.; $I^2 = \max(0, (Q - \text{df})/Q)$. Han m -values [Han and Eskin, 2012] use a Gaussian two-component mixture with a flat 0.5 prior: $P(\text{data} \mid \text{effect})$ at the RE pooled mean with variance $\text{SE}_i^2 + \tau^2$ versus $P(\text{data} \mid \text{no effect})$ at zero with variance SE_i^2 .

Classifier	$\bar{\theta}$	τ^2	Q	$p(Q)$	I^2	m_{THCA}	m_{SKCM}	m_{LGG}	m_{LUAD}	m_{COAD}
LogReg- ℓ_2	0.099	0.114	4538.8	$< 10^{-16}$	99.9%	1.00	0.10	0.02	0.04	0.03
RandomForest	0.065	0.017	184.1	$< 10^{-16}$	97.8%	1.00	0.22	0.06	0.16	0.10
XGBoost	0.001	0.000	0.12	0.998	0.0%	0.50	0.50	0.50	0.50	0.50
LogReg-elasticnet	0.099	0.110	4471.9	$< 10^{-16}$	99.9%	1.00	0.10	0.02	0.04	0.04
GradientBoosting	0.076	0.026	186.6	$< 10^{-16}$	97.9%	1.00	0.18	0.07	0.28	0.12

Table S1: Per-classifier random-effects meta-analysis of DIAL across five cancers. Four of five classifiers reject homogeneity overwhelmingly with THCA $m \geq 0.9999$; XGBoost is a null control.

Results. Four classifiers reject homogeneity with $I^2 > 97\%$ and $p(Q) < 10^{-16}$. In every rejected case THCA has $m \geq 0.9999$ while every other cohort sits at or below the Han $m \leq 0.1$ no-effect band. XGBoost is a negative control: Q is non-significant ($p = 0.998$), $I^2 = 0$, and every m -value collapses to the 0.5 flat prior — consistent with XGBoost never flipping in the primary analysis.

S4 Pathway-level DIAL (MSigDB Hallmark)

The v5.1 primary analysis measured DIAL on the 3,000 variance-top genes, where THCA BRAF-vs-RAS produced a 0.494 flip post-ComBat. The same 392 THCA samples, scored instead as a 50-dimensional MSigDB Hallmark [Liberzon et al., 2015] ssGSEA matrix [Barbie et al., 2009, Hänzelmann et al., 2013], return pooled DIAL = 0.000 across all five classifiers with $\text{AUC}_{\text{post}} \in [0.90, 0.98]$.

Classifier	DIAL gene	DIAL pathway	AUC_{post} gene	AUC_{post} pathway
LogReg- ℓ_2	0.494	0.000	0.006	0.964
LogReg-elasticnet	0.492	0.000	0.008	0.958
RandomForest	0.323	0.000	0.177	0.977
GradientBoosting	0.334	0.000	0.166	0.918
XGBoost	0.013	0.000	0.487	0.905

Table S2: Pooled gene-level versus pathway-level DIAL on THCA.

To distinguish aggregation-dilutes (some pathways flip but the pooled vote cancels) from pathway-level resilience (no pathway flips), we computed per-pathway LODO DIAL: one LogReg- ℓ_2 per Hallmark column, same joint 50-D ComBat. Zero of 50 pathways cross the batch-entangled threshold; median DIAL = 0.000, maximum = 0.085 (Oxidative Phosphorylation, the canonical RAS-high signature in thyroid carcinoma [Landa et al., 2016]). 29/50 pathways are clean `true.biology`, 13/50 `ambiguous`, 8/50 `no.signal`. The v5.1 3,000-gene DIAL 0.494 is therefore a high-dimensional residual-batch phenomenon, not a reversal of BRAF-vs-RAS biology.

When to worry vs when it is fine. A practical guide: (i) gene-level biomarker panel for clinical assay — watch gene-level DIAL on the training space; do not deploy under a flip; (ii) pathway enrichment — safe to proceed if every pathway has DIAL ≤ 0.1 , as here; (iii) deep-learning on raw expression — re-measure embedding-space DIAL after encoding.

S5 DE biomarker recomputation with DESeq2

Rationale. The upstream biomarker list used a Welch t -test on log2-TPM with BH-FDR. Reviewers prefer the counts-based NB-GLM in DESeq2 [Love et al., 2014, Muzellec et al., 2023], which provides size-factor normalisation, empirical-Bayes dispersion shrinkage, and a Wald test on moderated log2 fold-changes — all more robust for low-count genes than a t -test on TPM. We reran the BRAF-vs-RAS contrast with pydeseq2 v0.5.4.

Data and design. TCGA-THCA $n = 351$ (293 BRAF, 58 RAS). Design: `~subtype`. Cut: $\text{FDR}_{\text{BH}} < 0.05$, $|\log_2 \text{FC}| > 1$. Raw STAR counts were not on disk; integer pseudo-counts were derived from log2-TPM via $\text{round}((2^x - 1) \times 50)$ assuming a nominal 50M library. Size-factor normalisation absorbs scaling, but gene-level variance is understated relative to true counts; results are a reviewer-oriented sanity check rather than a definitive requantification.

Metric	Value
Old significant genes (prior list)	2,773
New significant genes (DESeq2)	6,283
Intersection	888
Old only	1,885
New only	5,395
Jaccard	0.109

Table S3: DE biomarker recomputation: Welch t vs DESeq2.

Results. All eight druggable targets from the prior biomarker slate (TACSTD2, TMPRSS4, PLEKHA6, CYP1B1, LDLR, GABRB2, B3GNT3, PTPRE) are retained under DESeq2 with p_{adj} ranging from 1.9×10^{-222} (TACSTD2) to 5.2×10^{-83} (LDLR) and $\log_2 \text{FC}$ from +2.58 to +5.29. No target needs to be reprioritised on the basis of this reanalysis.

S6 Quantum classifier DIAL comparison

Goal. Test whether the v5.1 THCA flip is classifier-paradigm specific. The primary analysis used linear and tree ensembles only. We add a third paradigm — quantum kernel / variational [Havlíček et al., 2019, Schuld and Killoran, 2019, Huang et al., 2021] — and re-measure DIAL on the same five cancers under LODO. If the THCA flip is task-intrinsic (a cohort-pairing property of the data) it should reproduce under quantum classifiers; if it is a peculiarity of linear models it should not.

Methods. Per-cancer, downsampled to $n \leq 80$ (stratified by $Y \times B$), PCA(8 components) to project to an 8-qubit Hilbert space. Three paradigms: classical SVC-RBF (baseline); QSVC (Qiskit-ML quantum kernel with ZZFeatureMap, reps=2); VQC (PennyLane variational classifier, 3 strongly entangling layers, 60 training iterations). DIAL was computed identically to the primary analysis.

Results. THCA under VQC flips from $\text{AUC}_{\text{pre}} = 0.673$ to $\text{AUC}_{\text{post}} = 0.114$ (DIAL 0.386), matching the magnitude of the classical flips. SVC-RBF agrees (DIAL 0.429). The THCA-specific direction inversion therefore extends from linear + tree to variational-quantum paradigms; the reviewer objection "DIAL is a linear-model weakness" is answered negatively. QSVC on THCA underfits ($\text{AUC} \approx 0.55$) — an underfit, not a flip. LGG/QSVC shows a sign-encoding artefact ($\text{AUC} \approx 0.18$ pre and post) but VQC and SVC-RBF on the same LGG folds both reach $\text{AUC} \geq 0.92$, so this is classifier-specific, not a batch flip.

Cancer	Family	AUC _{pre}	AUC _{post}	DIAL	Interpretation
THCA	QSVC	0.549	0.599	0.000	no_signal (underfit)
THCA	VQC	0.673	0.114	0.386	batch_entangled
THCA	SVC-RBF	1.000	0.071	0.429	batch_entangled
SKCM	QSVC	0.476	0.782	0.000	true_biology
SKCM	VQC	0.505	0.434	0.066	no_signal
SKCM	SVC-RBF	0.421	0.526	0.000	no_signal
LGG	QSVC	0.184	0.183	0.317	artefact (sign-inverted)
LGG	VQC	0.929	0.920	0.000	true_biology
LGG	SVC-RBF	0.998	1.000	0.000	true_biology
LUAD	SVC-RBF	0.774	0.847	0.000	true_biology
COAD	SVC-RBF	0.714	0.789	0.000	true_biology

Table S4: DIAL under QSVC, VQC, and SVC-RBF. THCA flips under VQC and SVC-RBF with magnitudes comparable to the classical LogReg/GBM flips.

S7 Han-lab-inspired methodology

Three Han-lab-style analyses strengthen the thyroid-specific claim: (S7.1) FastRNA-style per-cohort centering [Lee and Han, 2022], (S7.2) a BUHMBOX-inspired concept check for hidden sub-group heterogeneity [Han et al., 2016], and (S7.3) linear-mixed-model biomarker correction in the style of [Sul et al., 2018]. All use the harmonised THCA cohort (n=392, BRAF=321/RAS=71; TCGA-THCA=351/GSE27155=41; top-3000 variance genes).

S7.1 FastRNA-style cohort centering

FastRNA subtracts the per-cohort mean from every sample — variance is preserved, global location is removed, no empirical-Bayes step is required. Applied in place of ComBat and followed by the v5.1 LODO DIAL pipeline:

Classifier	AUC _{pre}	AUC _{post,centered}	DIAL _{centered}	DIAL _{ComBat}
LogReg- ℓ_2	0.994	0.993	0.000	0.494
LogReg-elasticnet	0.994	0.993	0.000	0.492
RandomForest	0.996	0.950	0.000	0.323
GradientBoosting	0.750	0.745	0.000	0.334
XGBoost	0.781	0.884	0.000	0.013

Table S5: FastRNA-style cohort centering eliminates the THCA flip.

ComBat LODO gave mean DIAL = 0.331 on THCA with two linear models at ≈ 0.49 (essentially perfect label polarity flips). FastRNA centering yields DIAL = 0.000 for every classifier. The THCA cross-cohort signal is recoverable once the (highly unbalanced) per-cohort mean is removed.

S7.2 BUHMBOX concept check

BUHMBOX [Han et al., 2016] detects whether shared heritability between two case groups reflects shared aetiology or hidden sub-group admixture. Our check is *conceptual*: for each subtype, we fit PCA on the pooled class-specific expression matrix and tested whether PC1 scores from TCGA-THCA and GSE27155 follow the same distribution (two-sample Kolmogorov–Smirnov). We observe KS = 1.00 with $p = 1.3 \times 10^{-40}$ for BRAF ($n = 28$ vs 293) and $p = 3.4 \times 10^{-14}$ for RAS ($n = 13$ vs 58); both flagged **hidden_heterogeneity**. PC1 aligns perfectly with cohort identity even after restricting to a single subtype.

S7.3 LMM biomarker correction

We fit per-gene `expression ~ subtype + (1 | cohort)` (`statsmodels.MixedLM.from_formula`, REML, L-BFGS) and compared to naive OLS `expression ~ subtype`. Both BH-FDR corrected across 3,000 top-variance genes. At $\text{FDR} < 0.05$ the naive test flagged 2,152 genes; LMM flagged 2,201. The lists overlap on 1,661 genes; 491 naive hits were lost with LMM and 540 were gained.

Status	n	median $ \beta $	median p_{naive}	median p_{LMM}	reading
same	1,969	0.559	—	—	strong-effect robust core
gained with LMM	540	0.265	3.1×10^{-1}	3.6×10^{-5}	rescued biology
lost with LMM	491	0.040	2.0×10^{-2}	4.2×10^{-1}	tiny-effect OLS artefacts

Table S6: LMM versus naive OLS biomarker lists on the 3,000-gene universe. LMM *rescues* more genes than it drops; dropped genes are tiny-effect cohort-confounded artefacts.

99.4% of the prior 2,773-gene biomarker slate that is testable by LMM retains significance under the cohort random-effect correction; all eight druggable targets remain LMM-significant.

S8 Direction-invariance lemma (full proof)

Setup. Work with standardised expression features $X \in \mathbb{R}^{n \times p}$, binary labels $Y \in \{0, 1\}^n$, and multi-valued batch indicator $B \in \{1, \dots, k\}^n$. Let $\mu_Y := \mathbb{E}[X \mid Y = 1] - \mathbb{E}[X \mid Y = 0] \in \mathbb{R}^p$ be the population biological direction and $\Delta_B := \{\mu_{B=b} - \mu_{B=b'} : b \neq b'\}$ the set of pair-wise batch-mean differences, with $\mathcal{V}_B = \text{span}(\Delta_B)$.

A *linear subtype-preserving batch-correction operator* is any affine map $P : \mathbb{R}^p \rightarrow \mathbb{R}^p$ whose sample-level action $P(X_i) = X_i - \hat{\gamma}_{B_i}$ satisfies (i) the post-correction mean within each batch is identical across batches and (ii) the within-label means $\mathbb{E}[P(X) \mid Y = c]$ preserve the pre-correction within-label means up to a label-independent constant. Constraints (i)–(ii) characterise the subtype-preserving ComBat family [Johnson et al., 2007] and its population limit.

Lemma 1 (First-moment annihilation). *Let P satisfy (i)–(ii) and assume $\mu_Y \in \mathcal{V}_B$. Then*

$$\mathbb{E}[P(X) \mid Y = 1] - \mathbb{E}[P(X) \mid Y = 0] = 0.$$

Proof. Constraint (i) implies, for every pair of batches $b \neq b'$, $\mathbb{E}[P(X) \mid B = b] = \mathbb{E}[P(X) \mid B = b']$. The batch-mean-difference vectors therefore lie in the kernel of P ; equivalently, the restriction $P|_{\mathcal{V}_B}$ is the zero map. Writing $P = \Pi_{\mathcal{V}_B^\perp} + r$ where r is the label-independent constant guaranteed by (ii), we have $P(\mu_Y) = \Pi_{\mathcal{V}_B^\perp}(\mu_Y) + 0 = 0$ under the hypothesis $\mu_Y \in \mathcal{V}_B$, and taking expectations conditional on Y yields the stated equality. $\square$

Corollary 1 (Residual predictive content). *Under the hypotheses of Lemma 1, the only expression direction on which the two classes can be separated in $P(X)$ is determined by the within-label second-moment residual*

$$\Sigma^\perp := \Pi_{\mathcal{V}_B^\perp}(\Sigma_1 - \Sigma_0)\Pi_{\mathcal{V}_B^\perp},$$

$\Sigma_c = \text{Cov}(X \mid Y = c)$. Any linear classifier f_{post} obtained by empirical risk minimisation with a strictly convex regulariser on $P(X)$ has its decision direction aligned with the leading eigenvector of $\Sigma^\perp$. Hence (i) $\text{auc}_{\text{flip}}(f_{\text{post}})$ may remain bounded away from 0.5 whenever $\Sigma^\perp$ is non-degenerate; (ii) the sign of the resulting discriminant is determined by spectral convention on $\Sigma^\perp$, not by μ_Y ; and (iii) $\text{DIAL}(f_{\text{post}}) > 0$ reports the batch-confounded predictive content that survives with its biological sign lost.

Sketch. Lemma 1 annihilates the first-moment label signal, so any linear separator on $P(X)$ must be recovered from the second-moment structure projected onto $\mathcal{V}_B^\perp$. Empirical risk minimisation over linear scoring rules with a strictly convex regulariser (the assumption f_{post} satisfies) returns a unique minimiser aligned with the leading eigenvector of the projected within-label covariance difference; the sign of that eigenvector is determined by spectral convention. See [Leek et al., 2010, Nygaard et al., 2016, Korsunsky et al., 2019] for empirical over-correction geometry that matches this residual-second-moment regime. $\square$

What the lemma does *not* prove. We are careful not to overclaim. The lemma states that the first-moment label signal is exactly annihilated when $\mu_Y \in \mathcal{V}_B$. It does not state, and we do not prove, a probabilistic AUC flip of the form $\text{AUC}(f_{\text{post}}) \rightarrow 1 - \text{AUC}(f_{\text{pre}})$ with probability one under some generating distribution. Such a statement would require the held-out-cohort data to be sampled from a distribution whose spectral gap between $\Sigma_1^\perp$ and $\Sigma_0^\perp$ is bounded below — a condition neither TCGA nor GEO are guaranteed to satisfy, and whose verification would itself require repeated-cohort simulation. The empirical five-cancer specificity pattern we report in §S3 is the substantive claim of the paper; the lemma+corollary pair provide the geometric reason why DIAL is the right thing to measure, not a distributional guarantee.

Remark on identifiability. When $\mu_Y \in \mathcal{V}_B$, a separate multi-class classifier trained to predict B from $P(X)$ will attain AUC close to chance because the batch means have been removed. The pathological regime is therefore the pair (high DIAL, low batch identifiability), which the primary analysis reports where the auxiliary classifier is fitable (three-or-more-cohort collections; not the present two-cohort LODO).

S9 Code and data availability

Code. All pipeline code is written in Python 3.11 and is available in the project repository under `notebooks_or_scripts/v5p1*.py` and `notebooks_or_scripts/v8*.py`. The harmonisation, ComBat, ComBat-seq, LODO, and meta-analysis steps are each stand-alone scripts with deterministic seeds.

Data. Inputs are downloaded at runtime from the GDC data portal and the NCBI GEO FTP server; no synthetic substitutes are used. TCGA mutation annotations were obtained from the MC3 public MAF. Per-cohort harmonised matrices and per-classifier LODO results are exported as TSV under `results/v5/` and `results/v8_statgen/`.

Dashboard. An interactive Plotly-based dashboard hosts every figure at gene and pathway level and is available at <http://40.82.129.113:8012/>.

Reproducibility. A `Dockerfile` and `docker-compose.yml` under the project root reproduce the full pipeline end-to-end from public sources. A frozen pip `requirements.txt` is shipped with the image.

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